

Abstract geometric lines in the top left corner, consisting of several overlapping, irregular polygons and lines in a light beige color, creating a complex, layered effect.

ROUTING IN ALL-OPTICAL LEO CONSTELLATIONS WITH HIGH-ALTITUDE GROUND STATIONS

Benoit Coeugnet

OUTLINE

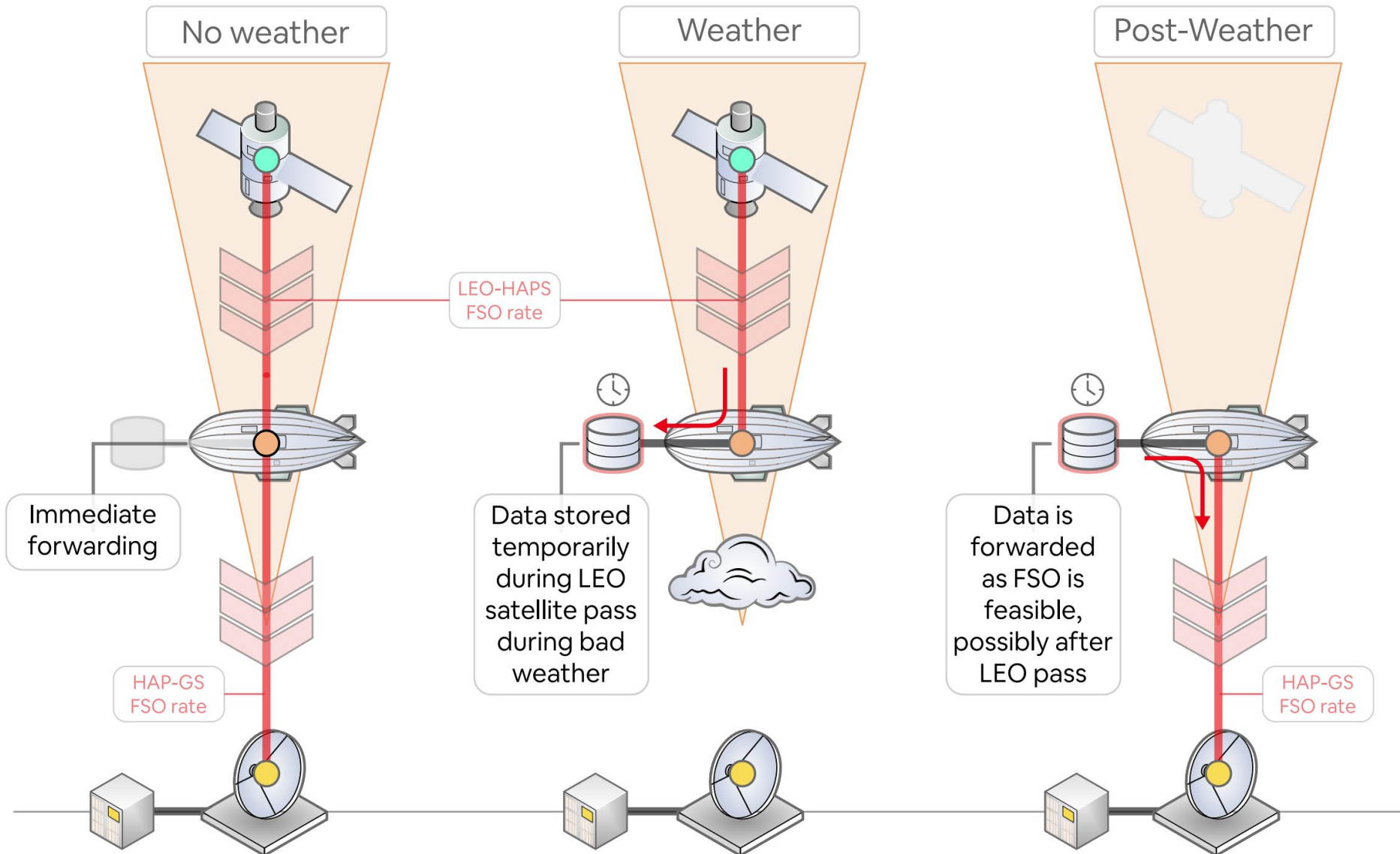
- **Motivation**
- **Architectures**
- **Routing solutions**
- **Simulation Parameters**
- **Results**
- **MDPs**

MOTIVATION

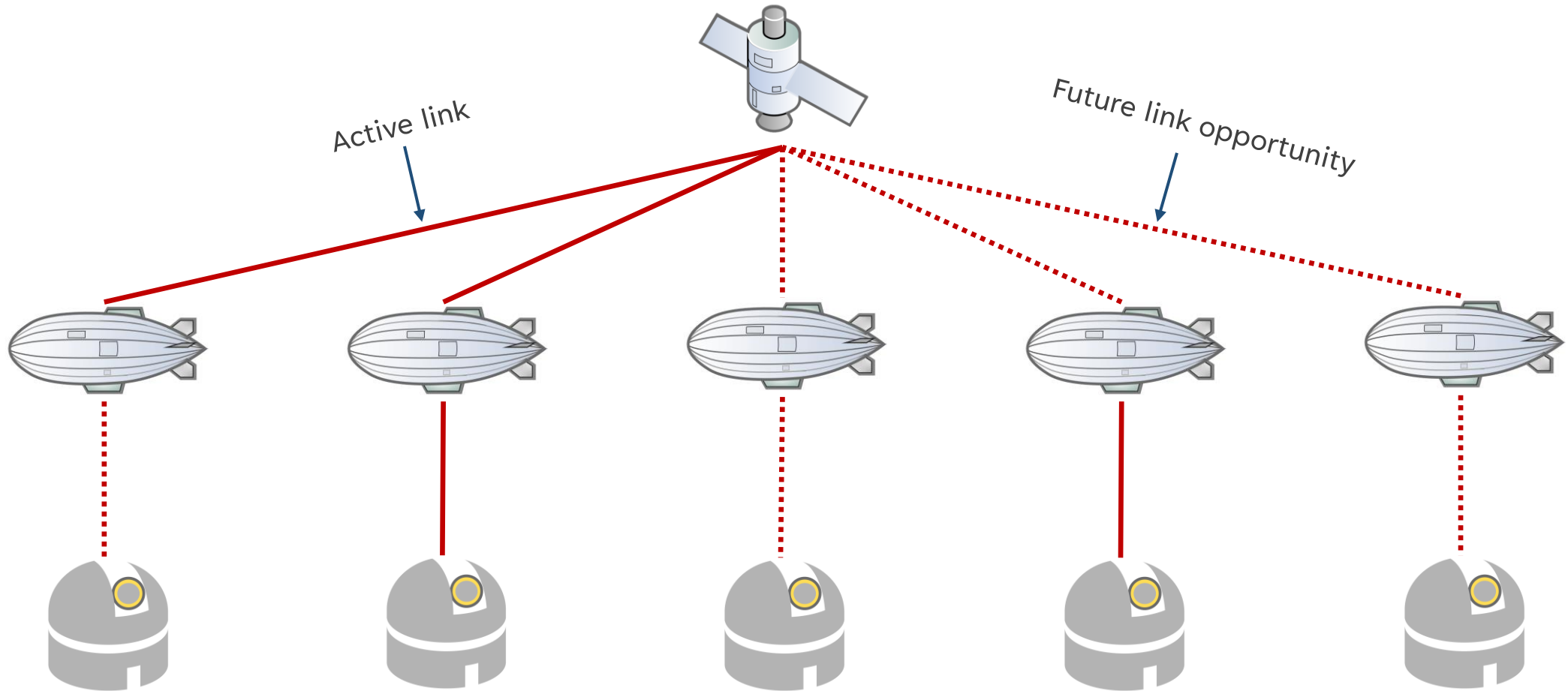
- The Copernicus Sentinels produce over 20,000 gigabytes of new data every day
- The RADARSAT and SAOCOM satellites use synthetic aperture radar (SAR) technology for maritime surveillance, disaster management, agriculture, and environmental monitoring



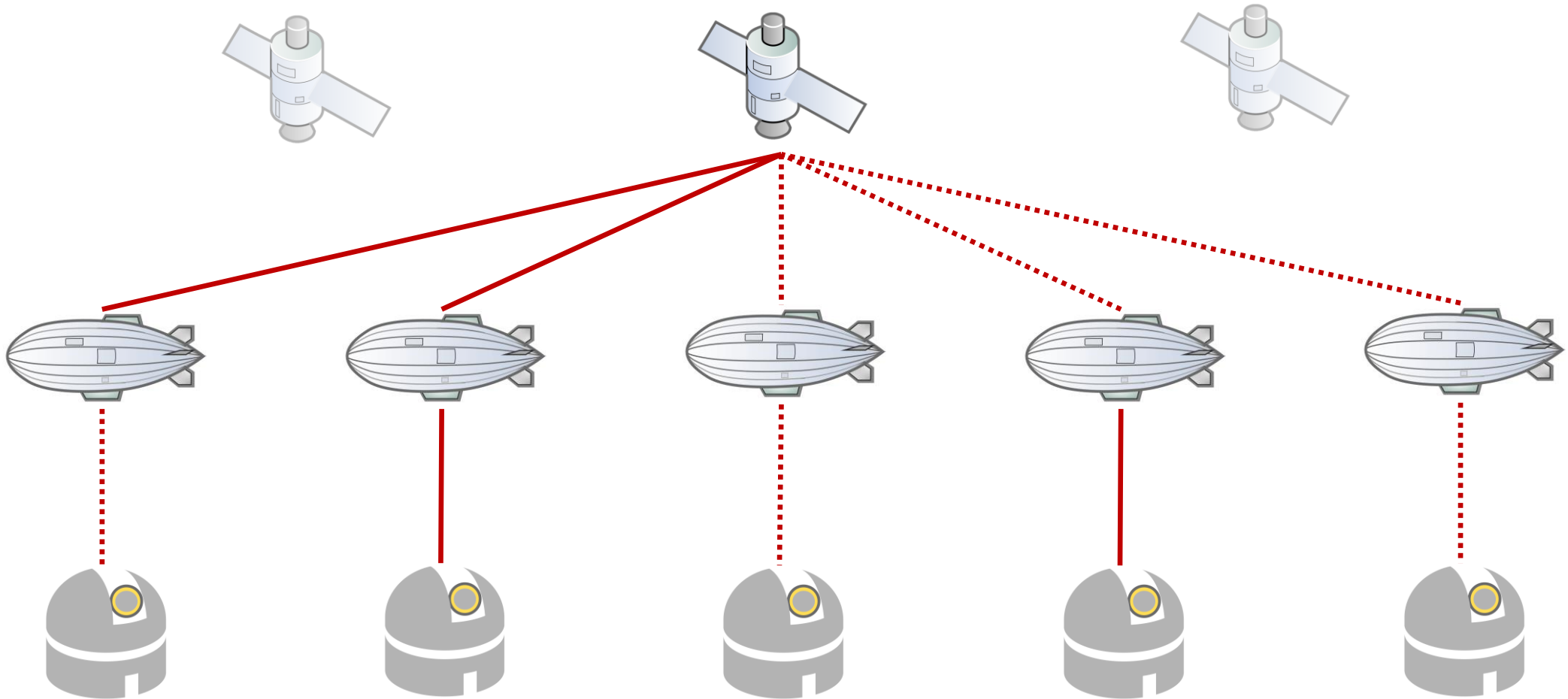
NOVEL ARCHITECTURE



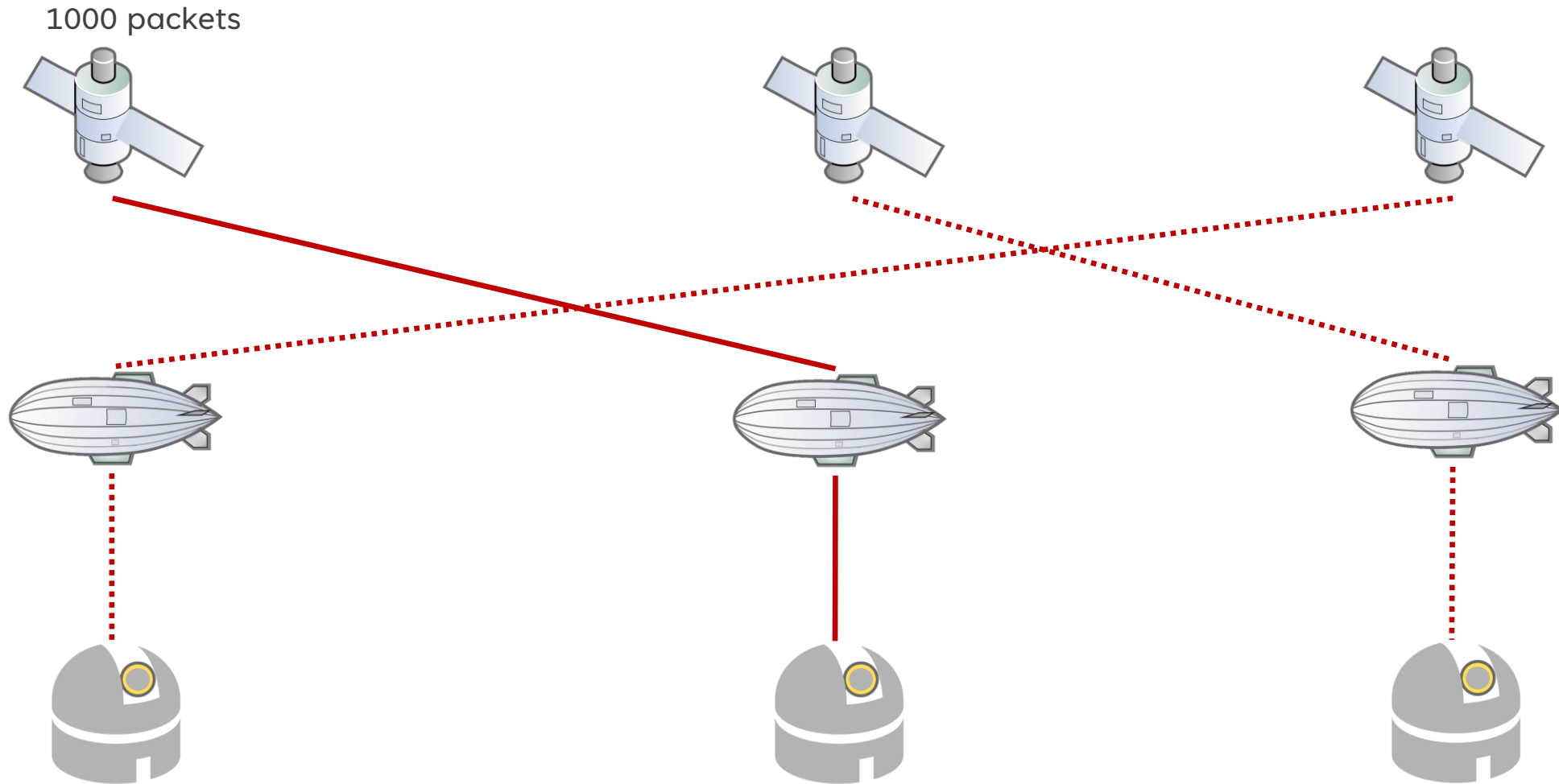
ONE LEO ARCHITECTURE



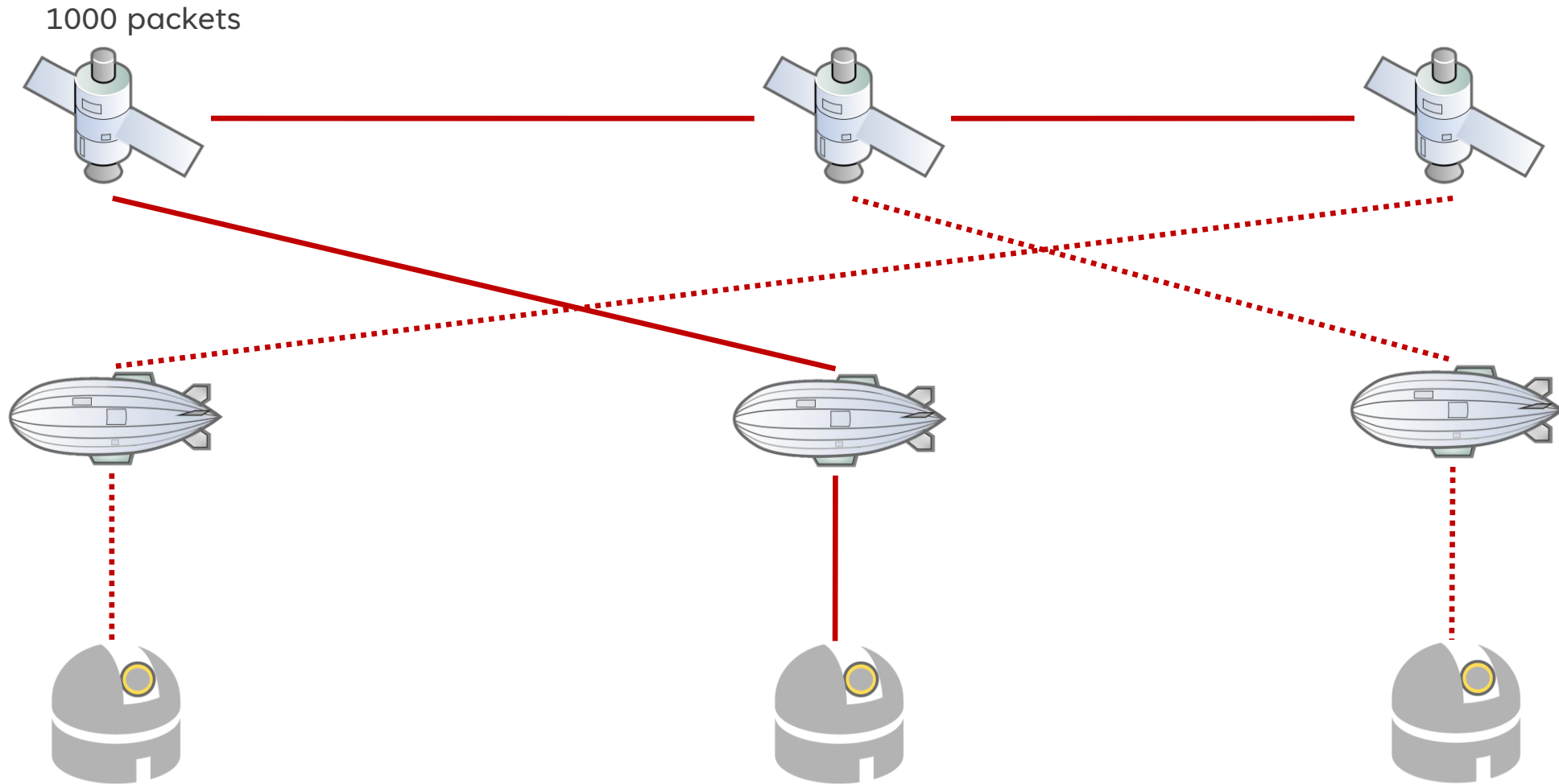
MULTIPLE LEOS ARCHITECTURE



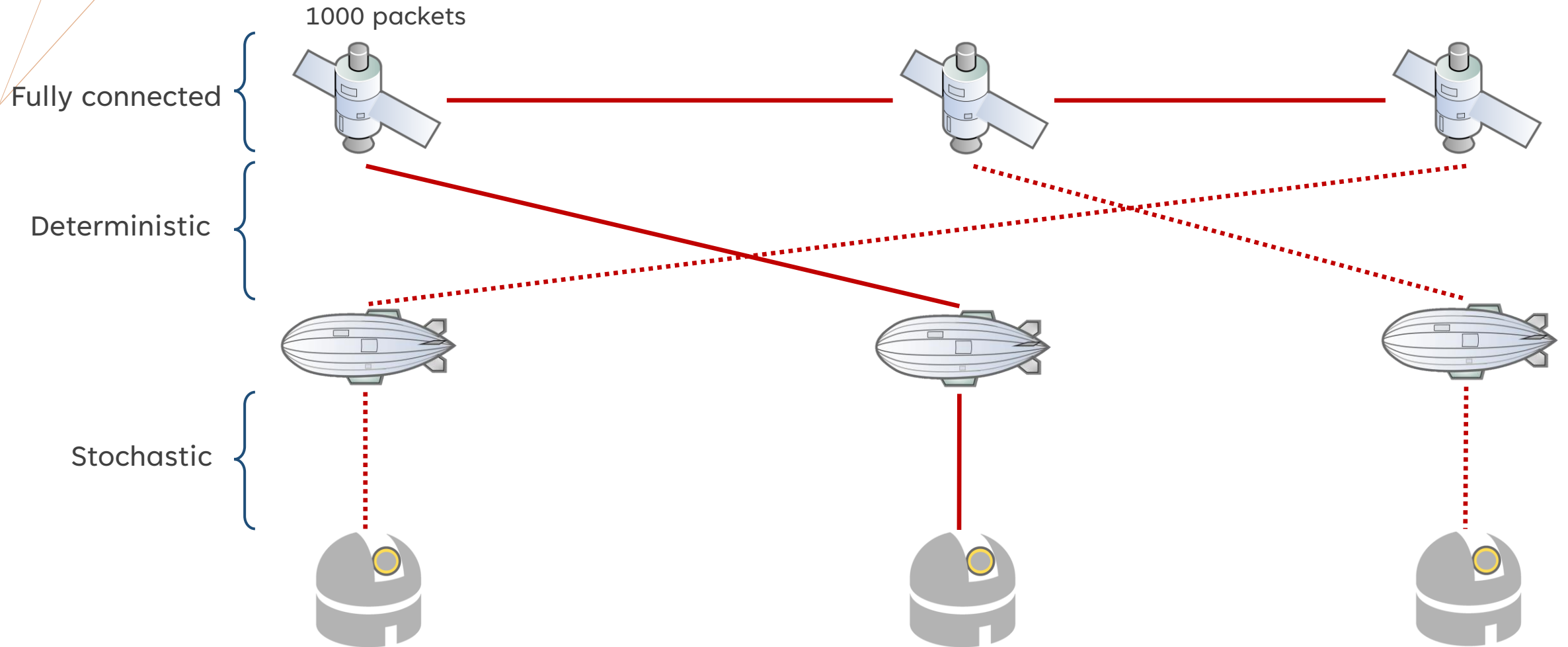
MULTIPLE LEOS ARCHITECTURE



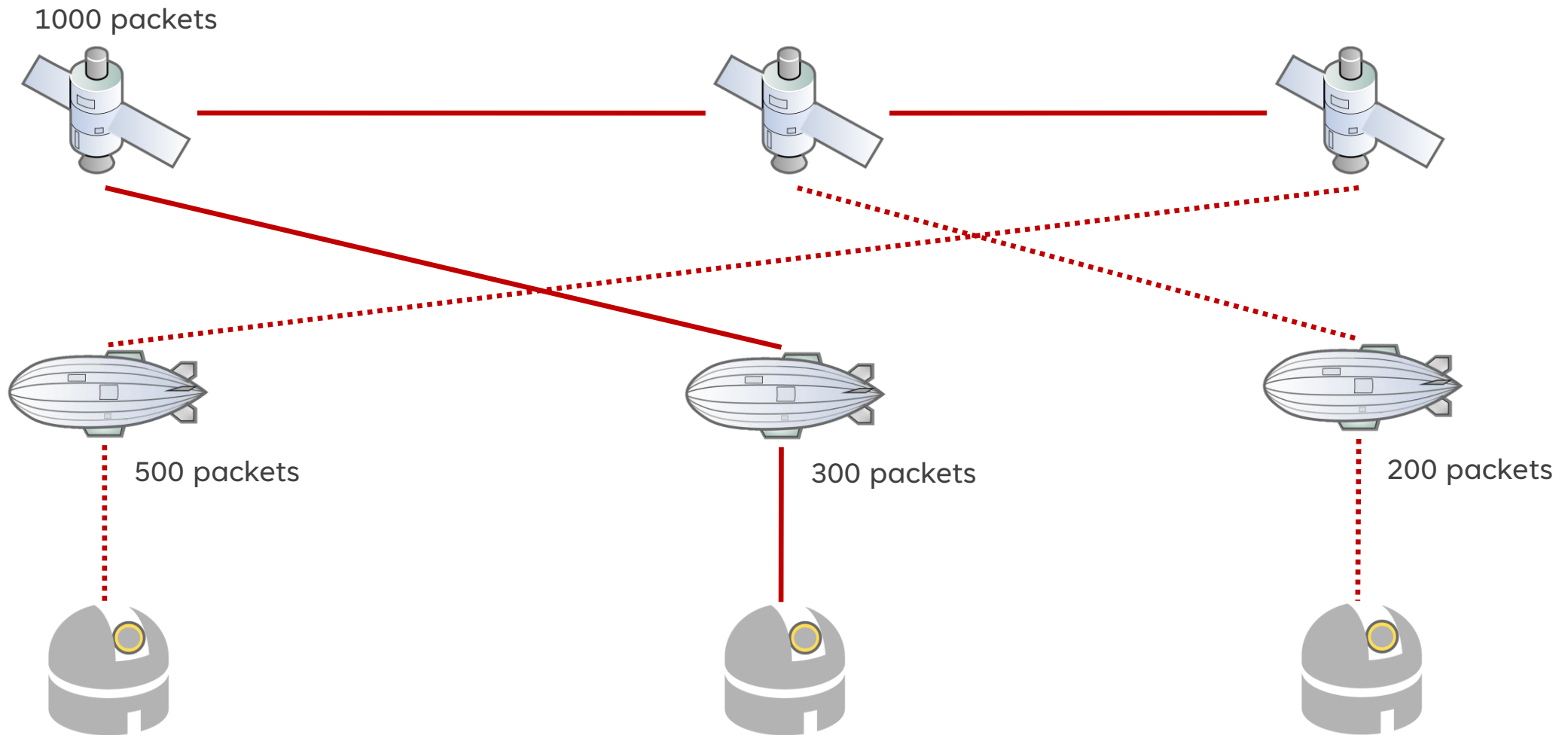
OISL LINKS



MULTIPLE LEOS ARCHITECTURE



GREEDY ROUTING



EQUAL ROUTING SOLUTION

Algorithm 2 Equal routing implementation

Require: *contactPlan*

for all bundles **do**

routes $\leftarrow$ Dijkstra's algorithm on *contactPlan*

for all *route* $\in$ *routes* **do**

if *route.node*[-2] = *bundle.number* % 5 **then**

if *route.arrivalTime* < *minArrivalTime* **then**

minArrivalTime $\leftarrow$ *route.arrivalTime*

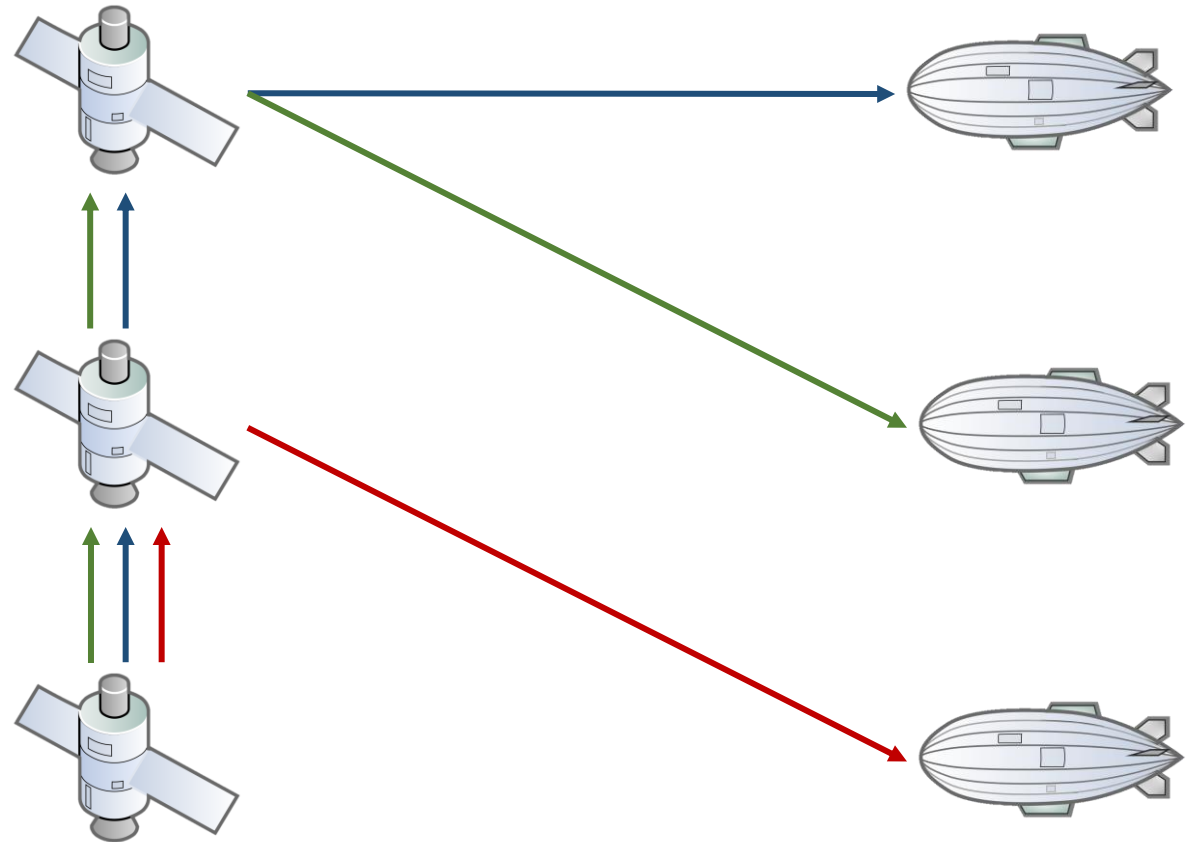
bestRoute $\leftarrow$ *route*

end if

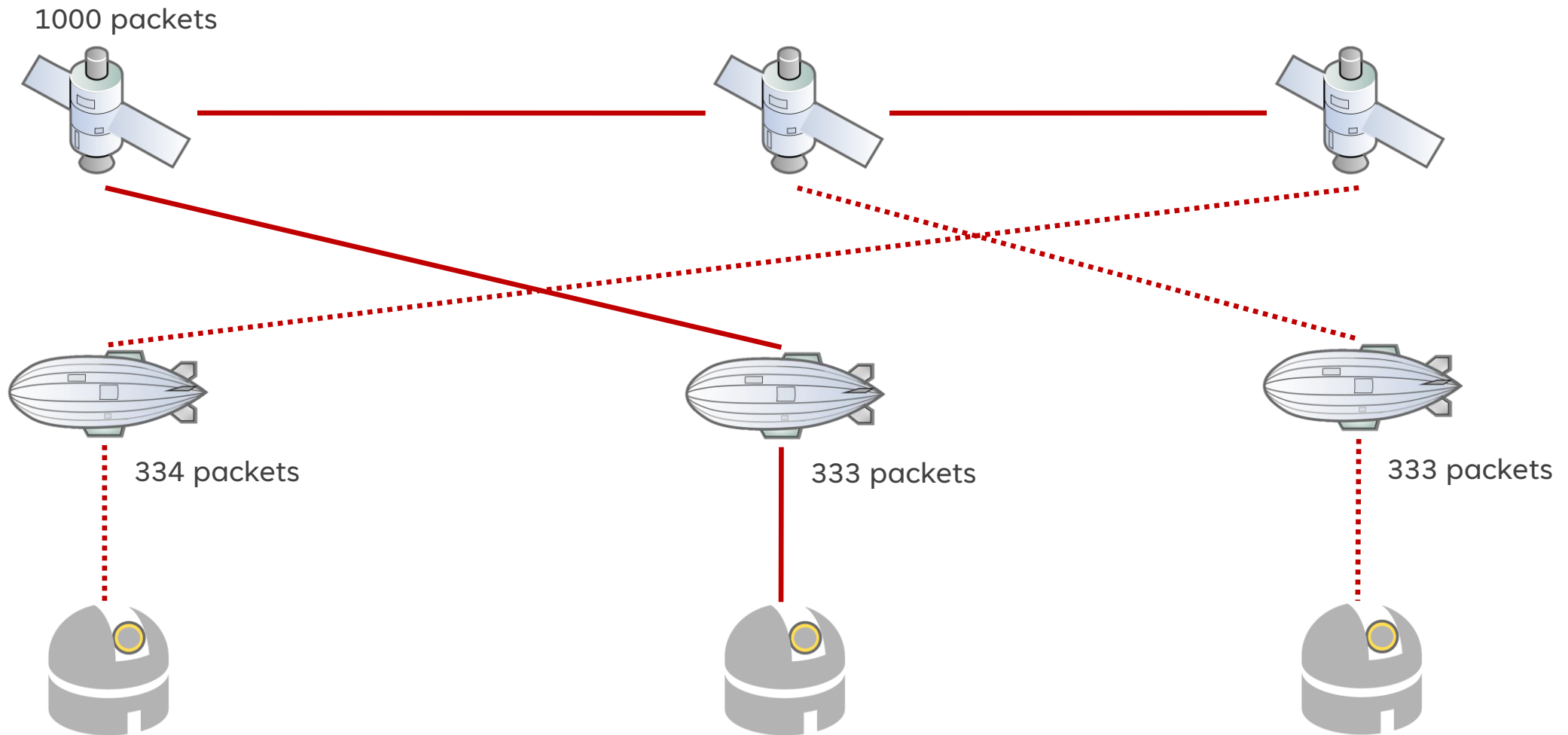
end if

end for

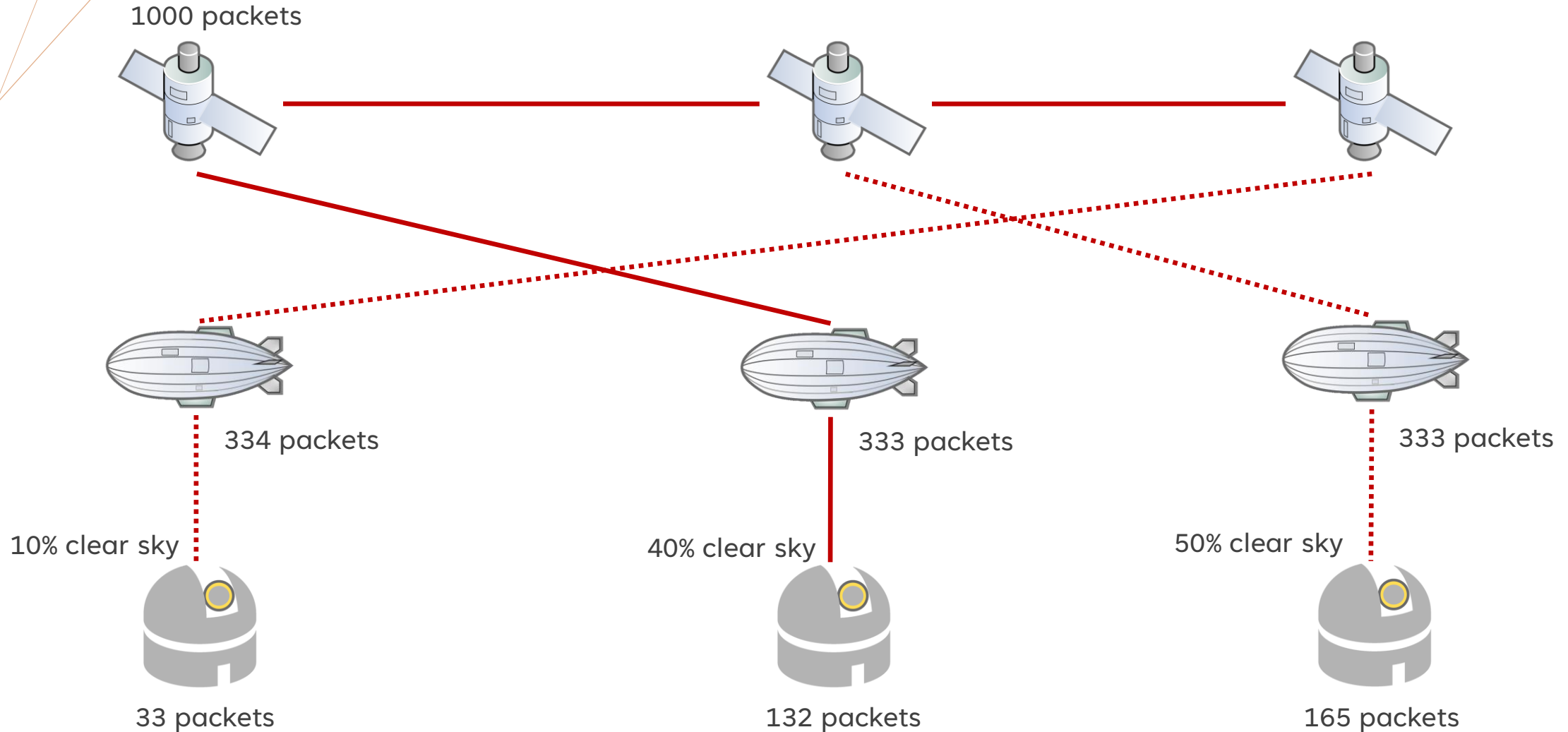
end for



EQUAL ROUTING



EQUAL ROUTING



WEIGHTED ROUTING SOLUTION

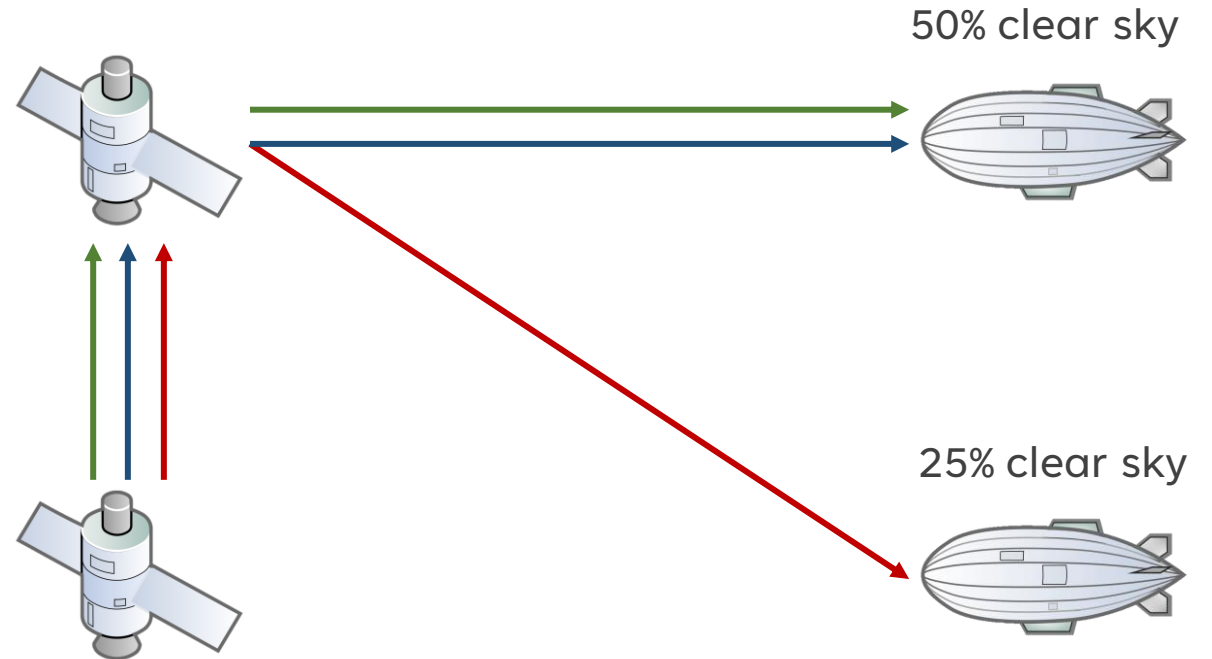
Algorithm 3 Weighted routing implementation

Require: *contactPlan*

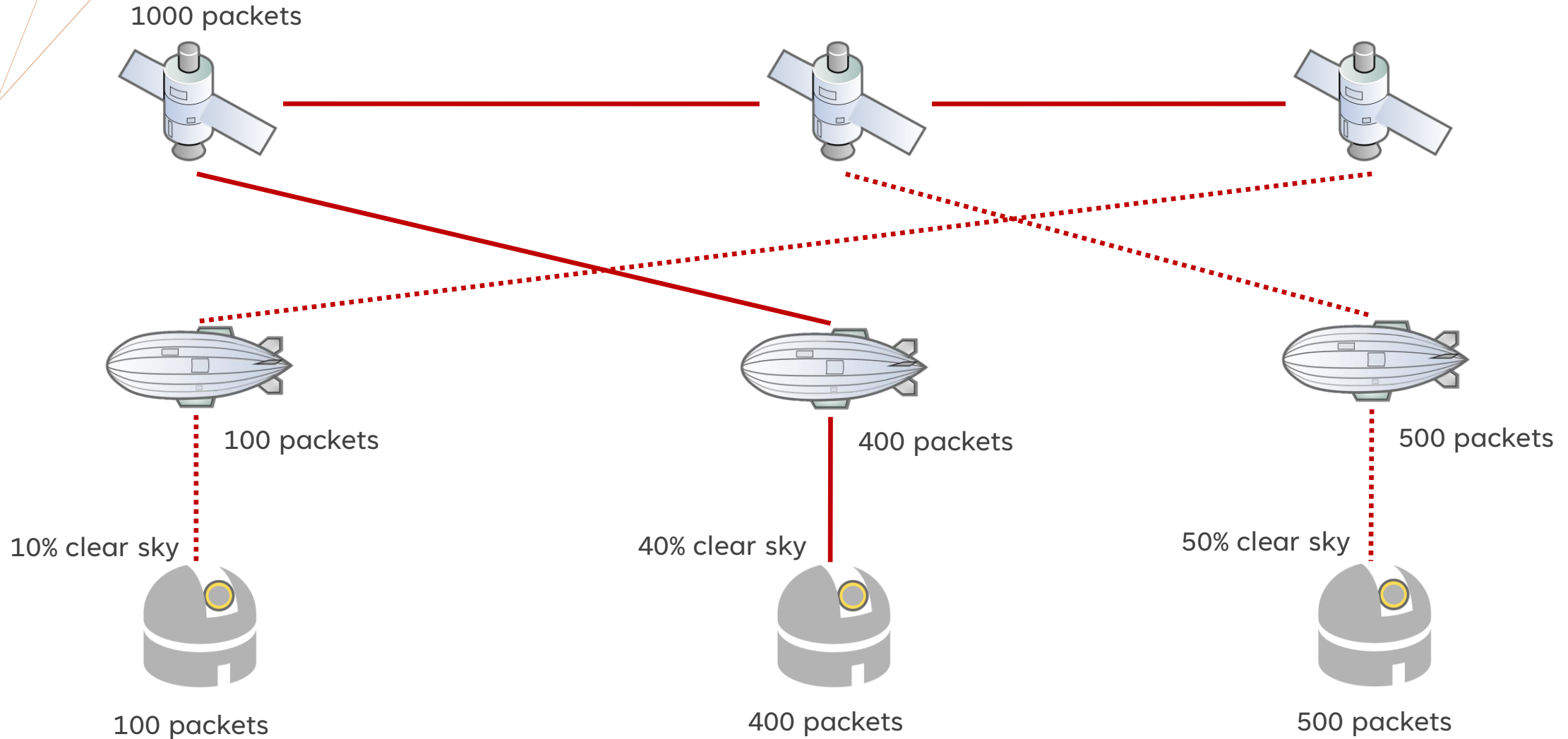
```

for all HAGS do
    cloudCoverh ← (ttrh / (ttrh + ttfh) × 100)
end for
for all HAGS do
    distributionh ← cloudCoverh / sum(cloudCover) × 100
end for
for all i < distribution.size() do
    for all j < distribution[i] do
        while hagsMapping[index] ≠ 0 do
            index ++
        end while
        hagsMapping[index] = hagsNumber
        index ← index + (100 / distribution[i])
    end for
    hagsNumber ++
end for
for all bundles do
    routes ← Dijkstra's algorithm on contactPlan
    for all route ∈ routes do
        hagsModulo ← bundle.number % 100
        if route.node[-2] = hagsMapping[hagsModulo] then
            if route.arrivalTime < minArrivalTime then
                minArrivalTime ← route.arrivalTime
                bestRoute ← route
            end if
        end if
    end for
end for

```



WEIGHTED ROUTING



PARAMETERS

GENERAL PARAMETERS

- **Local contact plan** as in reality
- Source routing : no re-routing

MODEL PARAMETERS

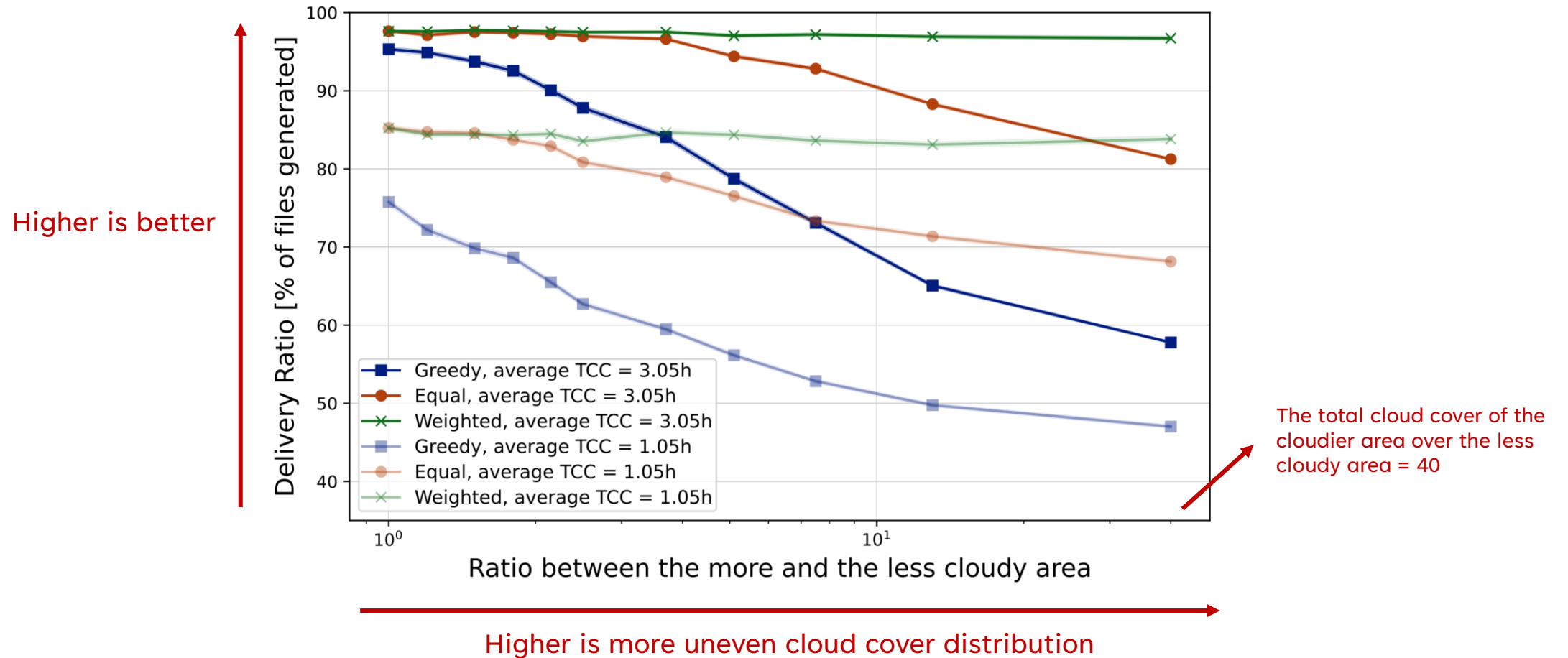
- 5 LEOs from Iridium Constellation
- 5 HAGS-GS from KSAT Ground Station Network

SIMULATION PARAMETERS

- 1000 bundles of 100 GB (for visual purpose)
- Data rate : 8 Gbps
- Time to Clear Sky (TCS) : 25h
- Time to Cloud Coverage (TCC) : variable to have different coverages

DELIVERY RATIO

Delivery Ratio : % of generated bundles that successfully reach any ground station



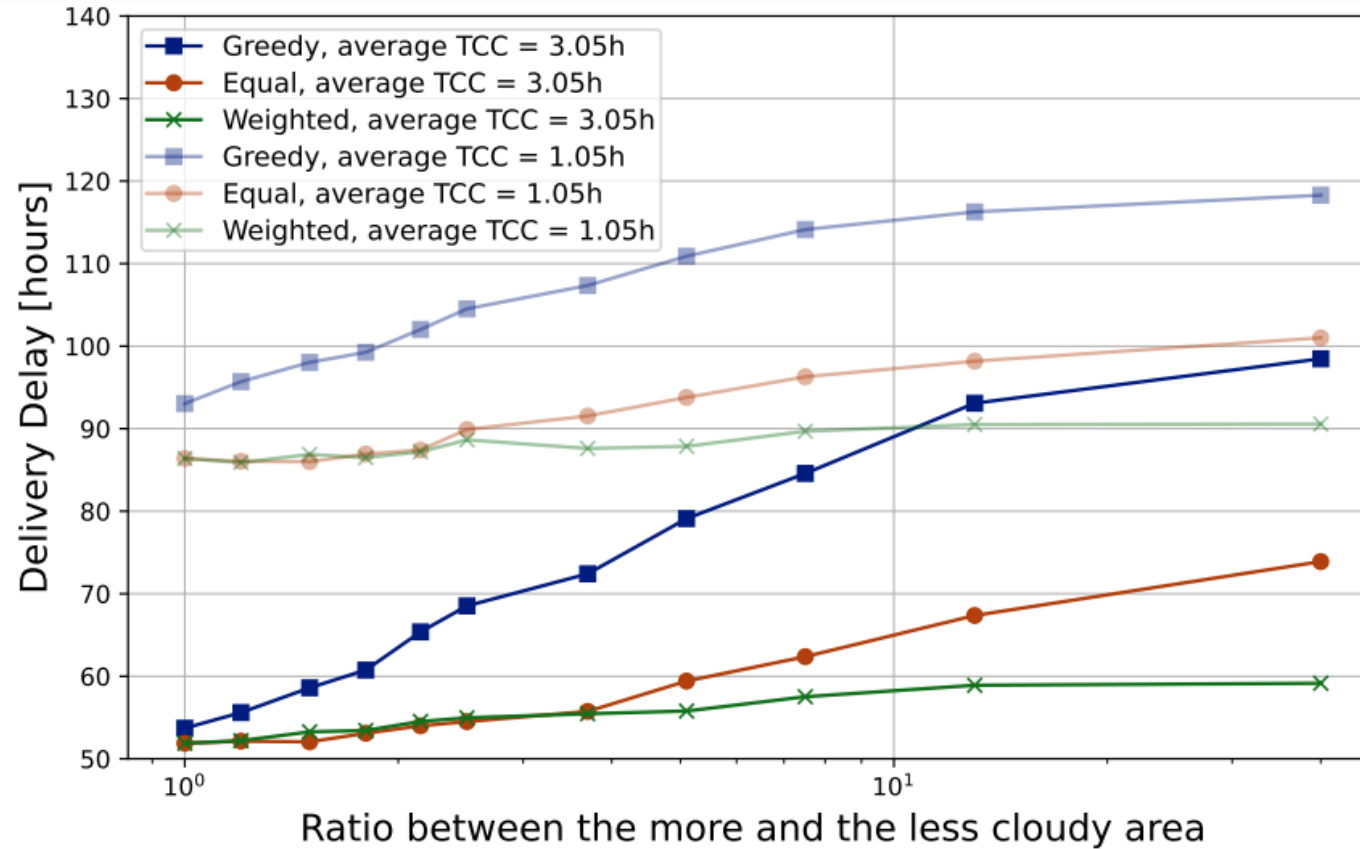
Distribution with a ratio of 40 and an average TCC of 1.05h :

0.05h / 0.55h / 1.05h / 1.55h / 2.05h

DELIVERY DELAY

Delivery Delay : average time of bundles that reach the destination

Less is better

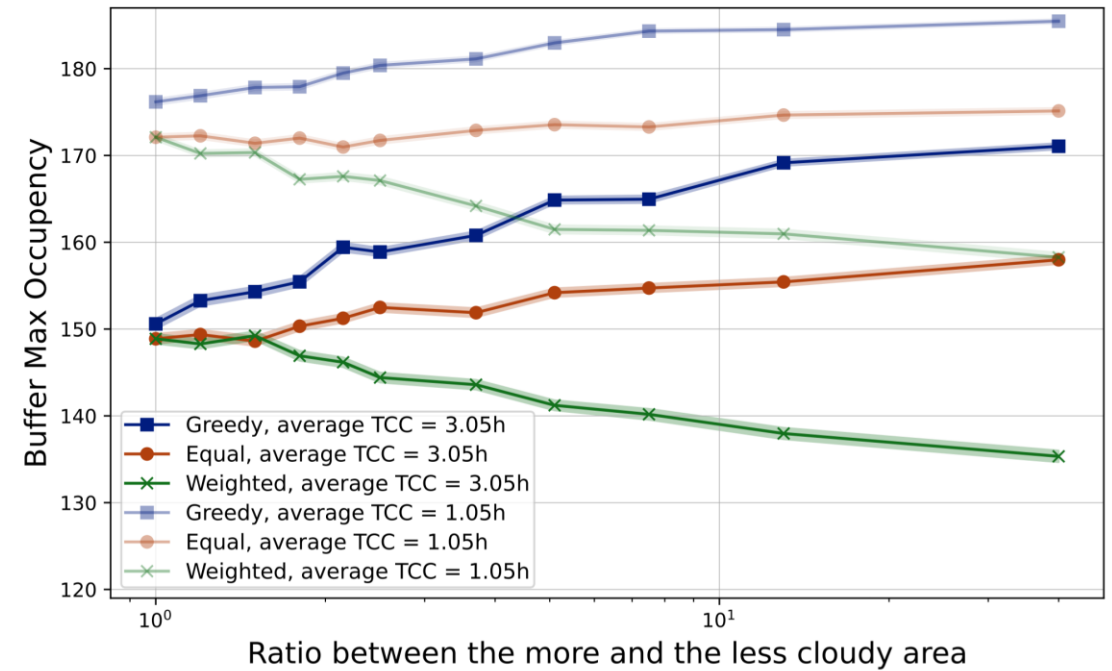
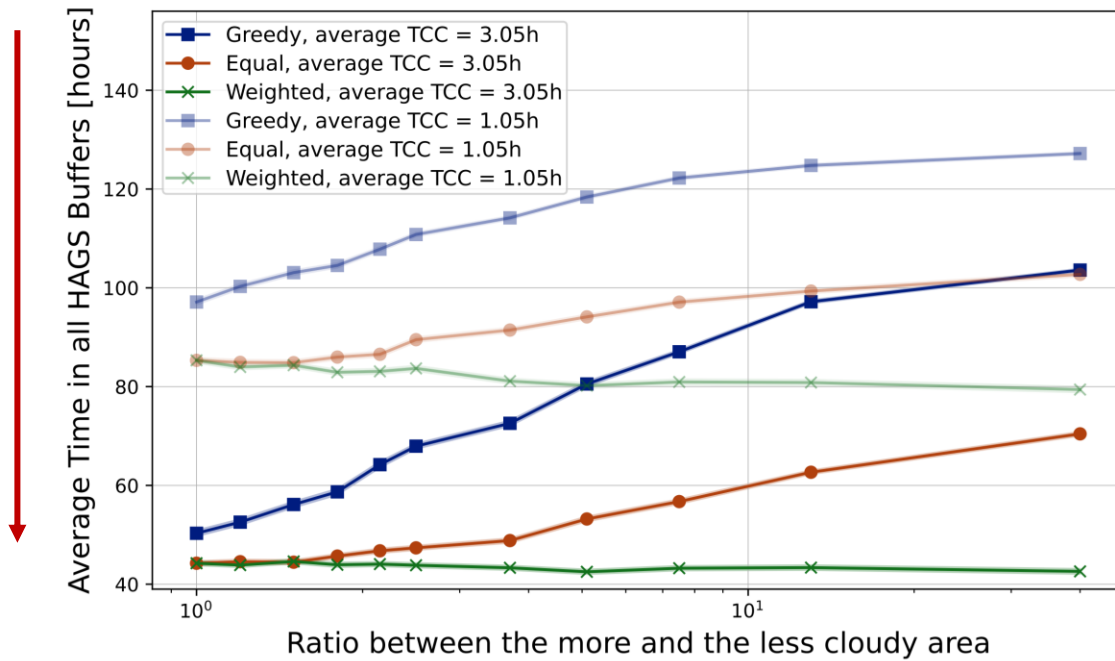


Higher is more uneven cloud cover distribution

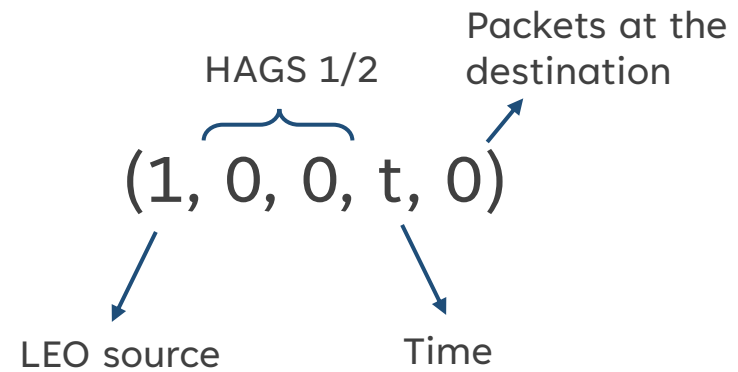
BUFFER OCCUPANCY

Max buffer occupancy : the maximum of bundles in all HAGS buffers

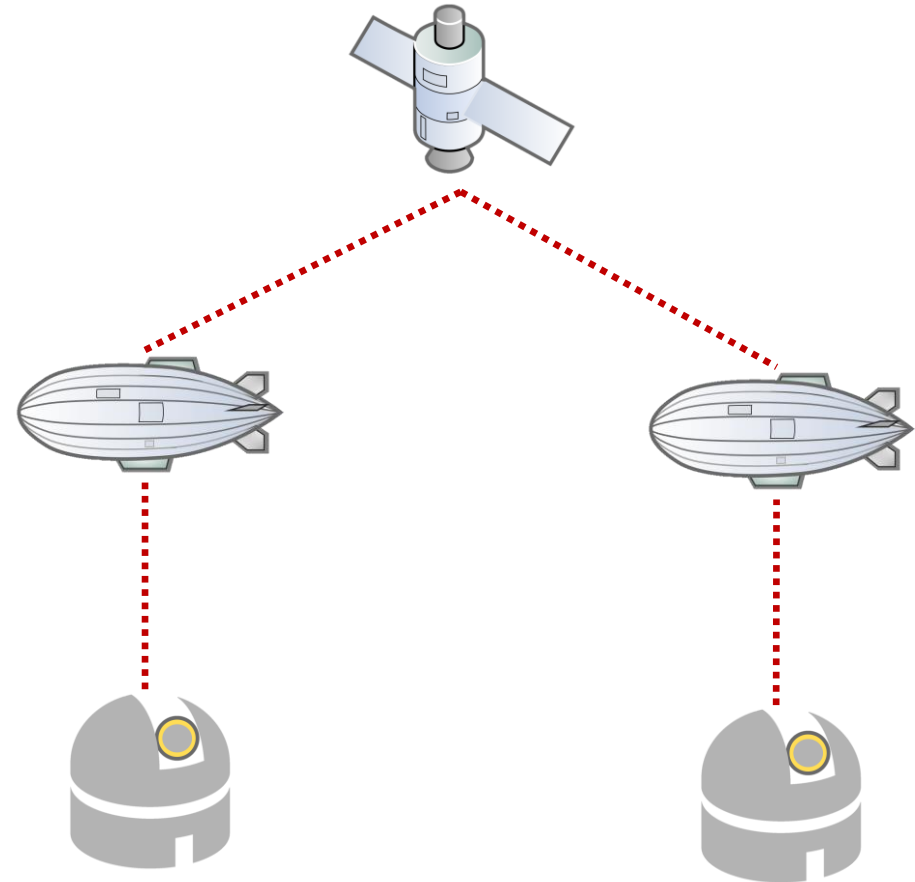
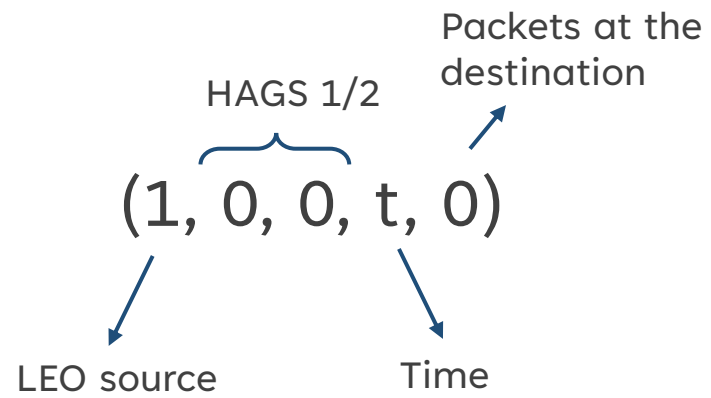
Less is better



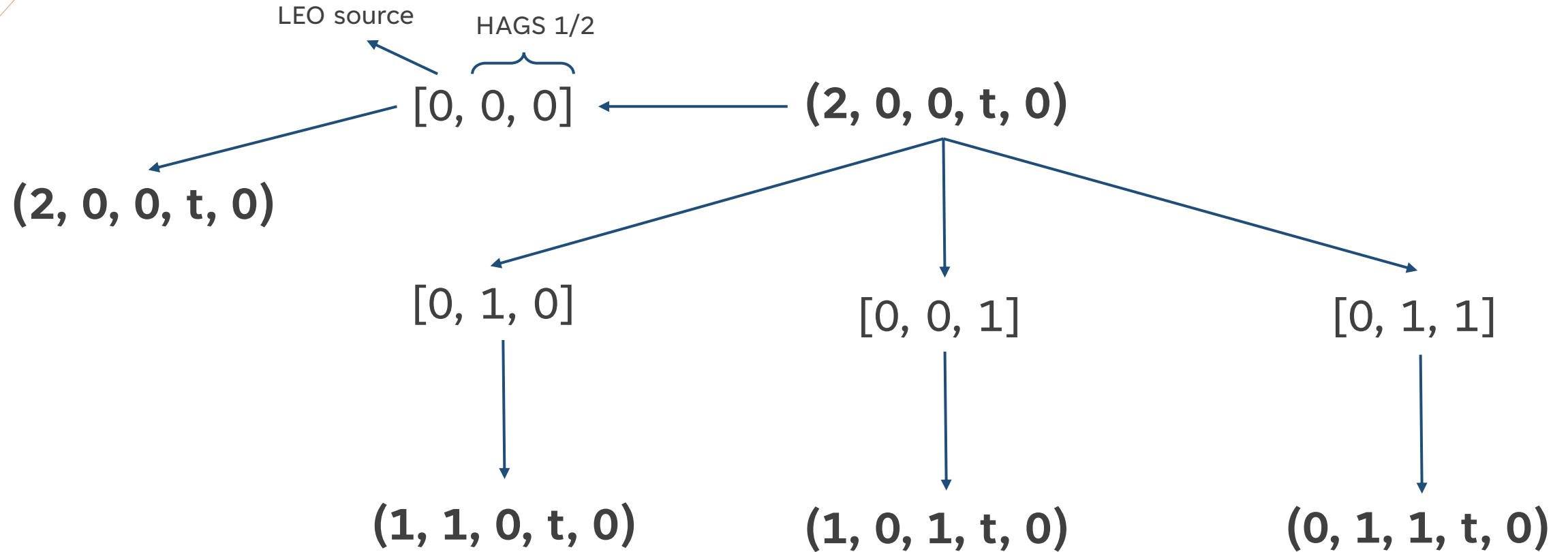
MARKOV DECISION PROCESS MODELISATION



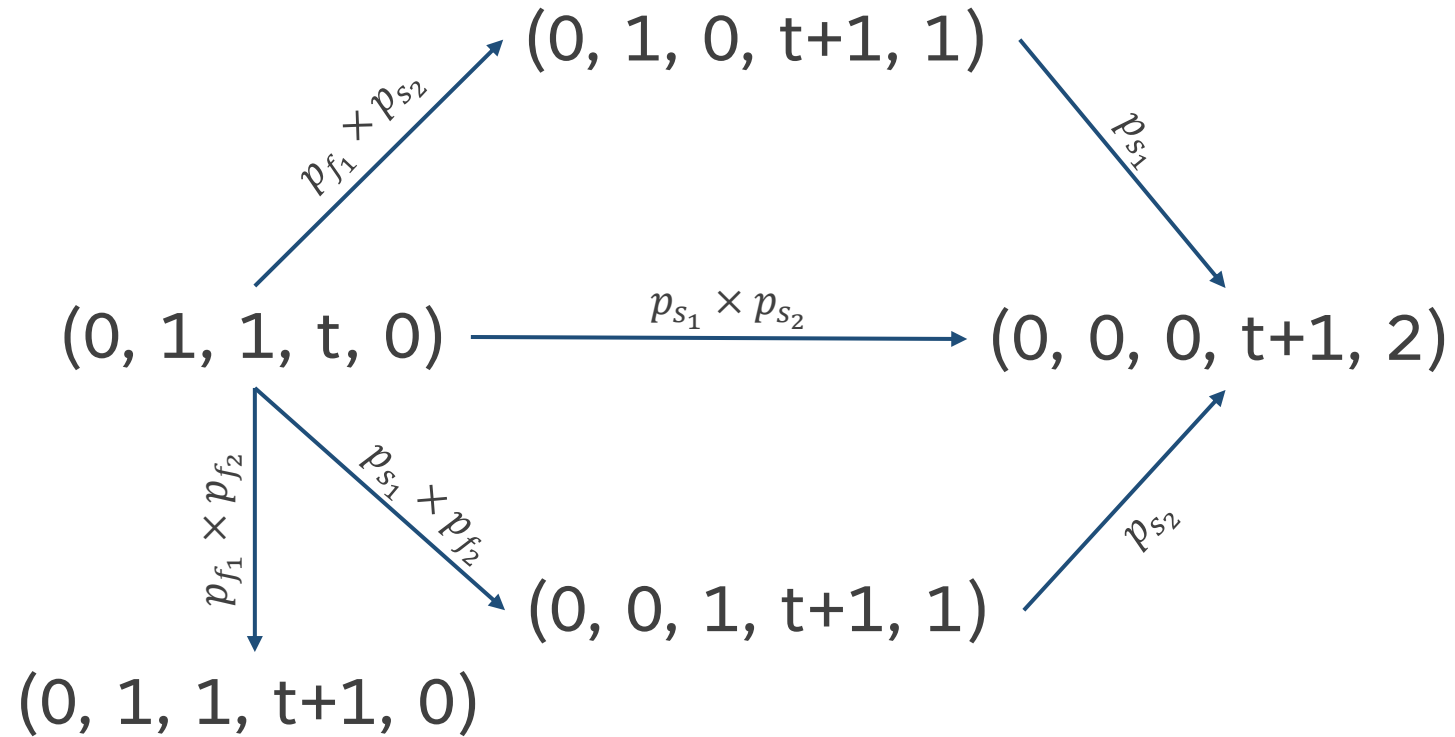
MARKOV DECISION PROCESS MODELISATION



MARKOV DECISION PROCESS MODELISATION



MARKOV DECISION PROCESS MODELISATION



A series of thin, light brown lines forming an abstract, overlapping geometric pattern on the left side of the slide. The lines intersect to create various polygonal shapes, some of which are filled with a very light brown color. The overall effect is a modern, minimalist design element.

THANK YOU !

TO GO FURTHER : 2 SOURCES

PARAMETERS : 500 bundles in the first LEO and 500 bundles in the fifth LEO

